

MID-STATE TECHNICAL COLLEGE • IT PROGRAMS

Managing Firewall Configurations

Unit 3 • Information Security 2 (10-151-111)



Learning Outcomes

By the end of this unit, you will be able to:



Distinguish running and candidate configurations

Explain what each holds, where it lives, and what a reboot does to uncommitted work.



Manage configuration files

Save, load, export, and import named snapshots, and revert to a known good state.



Commit changes deliberately

Preview, summarize, and validate a pending change before it touches the running config.



Coordinate work among administrators

Apply commit and config locks so concurrent admins do not overwrite each other.



View and filter firewall logs

Navigate the System and Configuration logs and build filters to isolate specific events.

LESSON 1

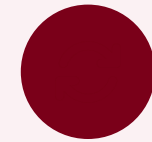
Running and Candidate



Two Configurations, One Firewall

A Palo Alto Networks firewall never applies a change the moment you click OK.

- The **running configuration** is what the firewall is enforcing right now. It lives on disk as `running-config.xml` and reloads at boot.
- The **candidate configuration** is the running config plus every change since the last commit. It lives in control-plane memory and affects no traffic.
- A **commit** validates the candidate and copies it into the running configuration, activating every pending change at once.



Clicking OK is not a change

Every dialog you close with **OK** writes to the candidate configuration only.

The web interface shows a pending-changes indicator at the top right — that is your reminder that nothing is live yet.

Where Each Configuration Lives

Three places a configuration can be, and only one of them is protected from a reboot:

Configuration	Where it lives	Survives reboot?	Active?
Candidate	Control-plane memory	No — unless saved	No
Saved snapshot	Firewall disk (.xml)	Yes	No
Running	Firewall disk + data plane	Yes	Yes

Save is not commit: Saving protects your work from a reboot. Committing makes it real. They are different buttons and they solve different problems.

Why the Distinction Bites People

- **Uncommitted changes are volatile.** The candidate lives in memory. A reboot before you commit or save takes the work with it.
- **Committing is not backing up.** A commit makes changes live; it does not hand you a file you can restore from later.
- **Commit is all-or-nothing by default.** A full commit activates every pending change in your accessible domain — including the one you made an hour ago and forgot.



SECURITY FOCUS

The candidate configuration is shared by every administrator logged in. Your commit can activate a colleague's half-finished rule change along with your own.

Why it matters: Nobody audits the change they did not know they were making.

LESSON 2

Managing Configuration Files



The Configuration Management Panel

Device > Setup > Operations — every configuration file operation lives on one panel.

Configuration Management

- Revert [Revert to last saved configuration](#)
[Revert to running configuration](#)
- Save [Save named configuration snapshot](#)
[Save candidate configuration](#)
- Load [Load named configuration snapshot](#)
[Load configuration version](#)
- Export [Export named configuration snapshot](#)
[Export configuration version](#)
[Export device state](#)
- Import [Import named configuration snapshot](#)
[Import device state](#)

- **Revert** — throw away candidate changes, back to the last save or to the running config.
- **Save** — write the candidate to a file on the firewall. Nothing is activated.
- **Load** — read a file back into the candidate. Still has to be committed.
- **Export / Import** — move a configuration file between the firewall and your workstation.

These operations are global in scope — a load or revert here affects the whole candidate configuration, not just your own edits.

Saving a Configuration

Two Save operations, and only one of them is safe to use twice.

- **Save candidate configuration** writes the candidate to the default snapshot file, `snapshot.xml`, overwriting whatever was there.
- **Save named configuration snapshot** writes it to a filename you choose, so previous snapshots survive. This is the backup operation.
- Neither one activates anything. A saved configuration is a file, not a running policy.

Save Named Configuration

Name

LIVE DEMO

Save a named snapshot before touching anything, then show it in the Load drop-down.

Loading a Configuration

Two ways back — one from a file you named, one from a version the firewall kept for you.

- **Load named configuration snapshot** reads a saved `.xml` into the candidate. You still have to commit it.
- **Load configuration version** loads one of the timestamped versions PAN-OS keeps automatically at every commit — useful when you never made your own snapshot.
- **Revert to last saved configuration** and **Revert to running configuration** discard candidate changes without loading a file.

Load Named Configuration

Name	
Decryption Key	FW-A-BaseConfig.xml
	FW-A-Enhanced-v1.xml

Load Versioned Configuration

Version	
Decryption Key	386 (2022/10/28 14:55:29+00)
	385 (2022/10/24 22:42:06+00)

Off the Firewall: Export and Import

A configuration stored only on the firewall is unavailable in exactly the emergency you saved it for.

Revert [Revert to last saved configuration](#)
[Revert to running configuration](#)

Save [Save named configuration snapshot](#)
[Save candidate configuration](#)

Load [Load named configuration snapshot](#)
[Load configuration version](#)

Export [Export named configuration snapshot](#)
[Export configuration version](#)
[Export device state](#)

Import [Import named configuration snapshot](#)
[Import device state](#)

- **Export named configuration snapshot** downloads a saved file from the firewall to your workstation.
- **Export configuration version** downloads one of the automatically retained committed versions.
- **Export device state** bundles the running config with licenses and certificates, for rebuilding a replacement unit.
- **Import named configuration snapshot** uploads a file from your workstation back onto the firewall.

Exports land in your browser's download folder — on the lab client, that is /home/lab-user/Downloads.

Naming Snapshots So They Help Later

A folder of files called test, test2, and final is not a set of restore points.

```
firewall-a-2026-09-08-pre-rule-cleanup  
firewall-a-2026-09-08-post-rule-cleanup  
firewall-b-2026-09-11-before-panos-upgrade
```

- Answer three questions in the filename: **which device**, **when**, and **why**.
- Snapshot **before** you start, not after. The point of a pre-change snapshot is that it captures the state you want to get back to.
- Pair the snapshot name with the commit description so the file and the audit trail agree.

LESSON 3

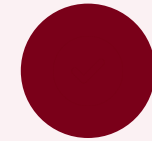
Committing Changes



What a Commit Actually Does

A commit is not a save button. It runs checks, then activates.

- PAN-OS runs **syntactic validation** — is the configuration well-formed — and **semantic validation** — does it actually make sense.
- Only if validation passes does the candidate become the running configuration.
- You can queue a commit while another is running; PAN-OS processes them in order rather than rejecting the second.
- **Auto-commits** are commits the firewall performs itself — at boot, or on an FQDN refresh — and they take priority in the queue.



Commit scope

A commit covers the **accessible domain** of the account you logged in with.

Your admin role decides how much of the firewall a full commit can reach.

Full Commit

Commit All Changes activates every pending change your account can reach — including other admins' work.

Commit

Doing a commit will overwrite the running configuration with the commit scope.

☒ Commit All Changes ☐ Commit Changes Made By: (1) admin

COMMIT SCOPE	LOCATION TYPE	OBJECT TYPE	ENTITIES	ADMINS
▶ policy-and-objects	Policy and Objects			
▶ device-and-network	Device and Network Configuration			

Preview Changes

Change Summary

Validate Commit

Note: This shows all the changes in login admin's accessible domain.

Description

Commit

Cancel

- The **Commit Scope** table shows which parts of the configuration have pending changes.
- Committing another administrator's changes requires the right permissions — and their knowledge.
- Use the **Description** field. Six months later the Configuration log tells you what changed; only the description tells anyone why.

Per-Admin Commit

Commit Changes Made By narrows the scope to selected administrators or locations.

Commit

?

Doing a commit will overwrite the running configuration with the commit scope.

☐ Commit All Changes

☒ Commit Changes Made By:(2) admin, admin-bob

COMMIT SCOPE	LOCATION TYPE	OBJECT TYPE	ENTITIES	ADMINS	INCLUDE IN COMMIT
▶ policy-and-objects	Policy and Objects				☒
▶ device-and-network	Device and Network Configuration				☒

Commit

?

Doing a commit will overwrite the running configuration with the commit scope.

☐ Commit All Changes

☒ Commit Changes Made By:(1) admin-bob

COMMIT SCOPE	LOCATION TYPE	OBJECT TYPE	ENTITIES	ADMINS	INCLUDE IN COMMIT
▶ policy-and-objects	Policy and Objects				☒

- One person's finished work can go live while another's half-built change stays pending.
- The **Include In Commit** column lets you drop individual scope rows even after filtering by admin.
- Your admin role decides whose changes you are allowed to see and commit here.

Reading the Commit Status Window

Closed it too early? Reopen it from the Tasks icon at the bottom right — the Task Manager keeps every job.

Commit Status

Operation

Commit

Status

Completed

Result

Successful

Details

Partial changes to commit: changes to configuration by administrators: admin
Changes to policy and objects configuration
Configuration committed successfully

Commit

Rule Shadow

Commit Status

Operation

Commit

Status

Completed

Result

Successful

Details

Partial changes to commit: changes to configuration by administrators: admin
Changes to policy and objects configuration
Configuration committed successfully

Commit

Rule Shadow

1 item

RULE	TYPE	COUNT
Allow-All-Traffic	security-rule	1

1 item

SHADOWED RULE

Rule 'Allow-All-Traffic' shadows rule 'Users_to_Internet'.

- **Warnings** do not stop the commit — which is exactly why they get ignored and should not be.
- **Errors** block the commit and leave the running configuration untouched until you fix them.
- **Shadowed rules** warn that a policy rule can never match because a broader rule above it already does.

Look Before You Commit

The Commit window offers three inspections. None of them touches the running configuration.



Link	What it shows	Question it answers
Preview Changes	Running vs. candidate, line by line	What exactly is different?
Change Summary	The individual settings in the commit	Which settings am I changing?
Validate Commit	Errors and warnings, without activating	Will this commit succeed?

Validation tells you the configuration is legal. Preview tells you it is what you meant.

Reading a Config Audit Diff

Preview Changes opens a Device Config Audit window: running on the left, candidate on the right.

Device Config Audit (firewall-a)

Mon Jun 14 21:34:53 UTC 2021

Administrators: admin

Legend: Added Modified Deleted

Local Device Changes	
Running Configuration	Candidate Configuration
352 }	352 }
353 update-server updates.paloaltonetworks.com;	353 update-server updates.paloaltonetworks.com;
354 update-schedule {	354 update-schedule {
355 threats {	355 threats {
356 recurring {	356 recurring {
357 weekly {	357 every-30-mins {
358 day-of-week wednesday;	358 at 10;
359 at 01:02;	359 action download-and-install;
360 action download-only;	360 }
	361 }
	362 }
	363 anti-virus {
	364 recurring {
	365 hourly {
	366 at 5;
	367 action download-and-install;

- **Green** — elements being added.
- **Yellow** — existing elements being modified.
- **Red** — existing elements being deleted.
- **Lines of Context** controls how many unchanged lines appear around each change. Default is 10.

Unexpected red is the single best reason to cancel a commit and look again.

Valid Is Not the Same as Correct

Validation is a syntax checker. It has no opinion about whether the change is the right one.

- A permitted-IP entry with `192.168.11.0/24` instead of `192.168.1.0/24` validates cleanly.
- It also locks every administrator out of the management interface the instant you commit it.
- Preview is the step that catches this, because a human reads the diff and notices the address is wrong.



SECURITY FOCUS

Every lockout incident in the field starts as a change that passed validation. The firewall enforced exactly what it was told.

Why it matters: The firewall cannot tell the difference between what you typed and what you meant.

LESSON 4

Multiple Administrators



Transaction Locks

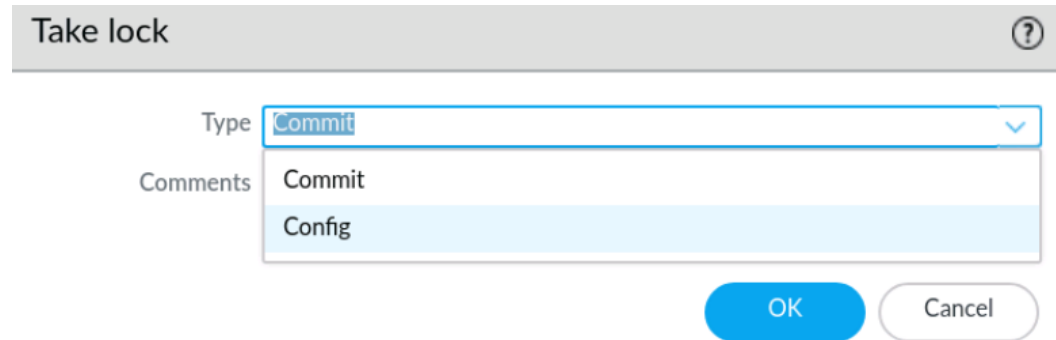
The candidate configuration is shared. Two locks keep concurrent administrators out of each other's work.

Commit Lock

- **Blocks other administrators from committing**
- Others can still edit the candidate configuration
- Nothing goes live until you release it or commit
- Use it when your change must activate as a complete set

Config Lock

- **Blocks other administrators from changing the candidate**
- The candidate is frozen except for you
- Prevents someone else's edit riding along with your commit
- Use it during a maintenance window or a large edit



A dialog box titled "Take lock" with a help icon (question mark) in the top right corner. It contains a "Type" dropdown menu with "Commit" selected. Below the dropdown is a "Comments" section with a text area containing "Commit" and "Config" on separate lines. At the bottom right are "OK" and "Cancel" buttons.

Take lock	
Type	Commit
Comments	Commit
	Config
OK Cancel	

[illegible]

- Only the administrator who set a lock can remove it — a superuser can clear it if that person is unavailable.
- Locks you hold are **released automatically when you commit**, which is convenient and also a common surprise.
- Fill in the **Comment** field. A lock with no explanation and no owner nearby stalls everyone else's work.

LESSON 5

Viewing and Filtering Logs



Accessing Firewall Logs

Monitor > Logs — same column controls, same filter bar, same export on every log.

PA-VM

DASHBOARDACC**MONITOR**POLICIESOBJECTSNETWORKDEVICE

Commit

Manual

Logs

- Traffic
- Threat
- URL Filtering
- WildFire Submissions
- Data Filtering
- HIP Match
- GlobalProtect
- IP-Tag
- User-ID
- Decryption
- Tunnel Inspection
- Configuration
- System**
- Alarms
- Authentication
- Unified
- Packet Capture
- App Scope
- Summary
- Change Monitor
- Threat Monitor
- Threat Map
- Network Monitor
- Traffic Map

RECEIVE TIME	TYPE	SEVERITY	EVENT	OBJECT	DESCRIPTION
03/17 15:45:32	general	informational	general		Connection to Update server closed: updates.paloaltonetworks.com, source: 192.168.1.254
03/17 15:45:15	url-filtering	high	url-cloud-connection-failure		CURL ERROR: Could not resolve host: pdx1prod.urlcloud.paloaltonetworks.com
03/17 15:44:55	url-filtering	medium	cloud-election		CLOUD ELECTION: cannot elect a cloud
03/17 15:43:55	url-filtering	high	url-cloud-connection-failure		CURL ERROR: Could not resolve host: s0000.urlcloud.paloaltonetworks.com
03/17 15:43:48	rtsg	medium	cloud-query-timeout	dns-signature	dns-signature cloud query timeout.
03/17 15:42:34	url-filtering	high	url-cloud-connection-failure		CURL ERROR: Could not resolve host: dl1prod.urlcloud.paloaltonetworks.com
03/17 15:42:13	url-filtering	medium	cloud-election		CLOUD ELECTION: cannot elect a cloud
03/17 15:41:13	url-filtering	high	url-cloud-connection-failure		CURL ERROR: Could not resolve host: s0000.urlcloud.paloaltonetworks.com
03/17 15:40:14	general	informational	general		Connection to Update server closed: updates.paloaltonetworks.com, source: 192.168.1.254
03/17 15:39:52	url-filtering	high	url-cloud-connection-failure		CURL ERROR: Could not resolve host: serverlist.urlcloud.paloaltonetworks.com
03/17 15:39:33	url-filtering	medium	cloud-election		CLOUD ELECTION: cannot elect a cloud
03/17 15:38:31	url-filtering	high	url-cloud-connection-failure		CURL ERROR: Could not resolve host: s0000.urlcloud.paloaltonetworks.com
03/17 15:37:10	url-filtering	high	url-cloud-connection-failure		CURL ERROR: Could not resolve host: serverlist.urlcloud.paloaltonetworks.com
03/17 15:36:50	url-filtering	medium	cloud-election		CLOUD ELECTION: cannot elect a cloud
03/17 15:35:50	url-filtering	high	url-cloud-connection-failure		CURL ERROR: Could not resolve host: s0000.urlcloud.paloaltonetworks.com
03/17 15:34:29	url-filtering	high	url-cloud-connection-failure		CURL ERROR: Could not resolve host: serverlist.urlcloud.paloaltonetworks.com
03/17 15:34:10	url-filtering	medium	url-cloud-connection-failure		Failed to open connection with the cloud after 250 consecutive tries.

1 2 3 4 5 6 7 8 9 10

Resolve hostname

Displaying logs 1 - 2020 per pageDESC

- **Traffic** — one entry per session, with the rule that allowed or blocked it.
- **Threat** — detections from Security profiles.
- **System** — what the firewall did on its own.
- **Configuration** — what administrators did to the firewall.
- **Unified** — Traffic, Threat, URL, WildFire, and Data Filtering in one view.

System Log vs. Configuration Log

Two logs, two different stories about the same change.

System Log

- **What the firewall did on its own**
- Service starts, NTP sync, update downloads
- Commit and load job results
- Severity: `critical high medium low informational`

RECEIVE TIME	ADMINISTRAT...	HOST	CLIENT	COMM...	RESULT	CONFIGURATION PATH	FULL PATH
03/03 20:39:51	admin	192.168.1.20	Web	commit	Submitted		
03/03 20:35:00	admin	192.168.1.20	Web	set	Succeeded	deviceconfig system	/config/devices/entry[@name='localhost.localdo...
03/03 20:34:18	admin	192.168.1.20	Web	edit	Succeeded	vsys vsys1 rulebase security rules Block-Bad-IPs-Outbound	/config/devices/entry[@name='localhost.localdo... Bad-IPs-Outbound']
03/03 20:32:36	admin	192.168.1.20	Web	commit	Submitted		
03/03 20:32:19	admin	192.168.1.20	Web	move	Succeeded	vsys vsys1 rulebase security rules Block-Bad-IPs-Inbound	/config/devices/entry[@name='localhost.localdo... Bad-IPs-Inbound']
03/03 20:32:11	admin	192.168.1.20	Web	edit	Succeeded	vsys vsys1 rulebase nat rules Extranet_to_Internet	/config/devices/entry[@name='localhost.localdo...

Configuration Log

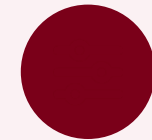
- **What administrators did to the firewall**
- Every change to the configuration
- Who made it, when, and from which client address
- The audit trail an incident review will ask for

RECEIVE TIME	TYPE	SEVERITY	EVENT	OBJECT	DESCRIPTION
03/03 20:41:34	general	informational	general		User admin accessed tab: monitor
03/03 20:41:12	vpn	informational	ike-nego-p1-fail-common	192.168.50.10[500]	IKE phase-1 negotiation is failed. Couldn't find configuration for IKE phase-1 request for peer IP 192.168.50.10[500].
03/03 20:40:53	general	informational	general		Commit job succeeded. Completion time=2022/03/03 20:40:54. Jobid=15. User:admin
03/03 20:40:49	vpn	informational	ike-nego-p1-fail-common	192.168.50.10[500]	IKE phase-1 negotiation is failed. Couldn't find configuration for IKE phase-1 request for peer IP 192.168.50.10[500].
03/03 20:40:38	general	informational	general		Config installed
03/03 20:40:36	vpn	informational	ike-nego-p1-fail-common	192.168.50.10[500]	IKE phase-1 negotiation is failed. Couldn't find configuration for IKE phase-1 request for peer IP 192.168.50.10[500].

Making a Log Table Readable

Before you filter, shape the table so the column you care about is the one you see first.

- The small **drop-down arrow** in any column header opens **Columns** — hide what you do not need, reveal what is hidden by default.
- **Drag a column header** to move it. Pulling Severity to the far left makes a noisy System log readable at a glance.
- **Export to CSV** writes the current view to a file — the thing you attach to a ticket or hand to an auditor.



Column changes are per-user

Hiding and reordering columns changes your view of the log, not the log itself. Nothing is deleted and no other administrator is affected.

LIVE DEMO

Hide the Object column, then drag Severity to the far left of the System log.

Building Filters by Clicking

Click a value in any column and the interface writes the filter syntax for you.

```
( severity eq informational )  
( severity eq informational ) and ( subtype eq general )
```

- Clicking `informational` in the Severity column produces the first filter. Clicking `general` in the Type column joins it with `and`.
- Typing into the filter bar does not run the query — the **Apply Filter** arrow does.
- **Clear Filter** before you leave a log. Otherwise the next person to open it sees a partial view and does not realize it.

The Filter Builder

For time ranges, negation, and comparisons — the plus icon at the right of the filter bar.

Add Log Filter

(addr.src in 192.168.1.25) and (addr.dst in 192.168.50.53)

Connector	Attribute	Operator	Value
and	Count	in	192.168.50.53
or	Destination Address	not in	
	Destination Category		
	Destination Country		
	Destination Dynamic Address Group		
	Destination FDI		

☐ Negate

Add Apply Close

- **Connector** — *and* or *or*, plus a Negate checkbox.
- **Attribute** — the field to test: Severity, Event, Object, Time Generated.
- **Operator** — equal, not equal, greater than or equal, less than or equal, in.
- **Value** — for time attributes, a date picker and a time picker.

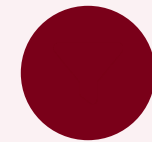
Add, then Apply: Each condition needs Add to append it to the filter string. Forgetting Add is the most common reason a Filter Builder session produces nothing.

Filtering by Time

The most useful filter in an investigation is almost always a time window.

```
( severity eq informational ) and ( time_generated geq '2026/09/08 14:25:00' )
```

- Read the current time from **Dashboard > General Information** first, then subtract your window.
- **Time Generated** is when the event happened. **Receive Time** is when the log system recorded it.
- On a single firewall the two are nearly identical — they diverge once logs are forwarded, and that difference matters in an investigation.



Same everywhere

The Filter Builder and this syntax work in every log table on the firewall — Traffic and Threat included. Learn it once here.

Security Focus: Misconfiguration Is the Threat



SECURITY FOCUS

Gartner's widely cited finding is that roughly 99% of firewall breaches trace to misconfiguration rather than product flaws. The device works; the rules on it do not. An overly broad rule, a management interface reachable from the wrong network, a change nobody reviewed — each is a perfectly valid configuration that a firewall will happily enforce.

Why it matters: Preview, validate, and revert are the controls that let a human catch the difference between valid and correct.

NIST SP 800-128 covers the other half — defining a secure baseline, approving changes against it, and monitoring for drift.

Change Discipline in Six Habits

The firewall gives you the tools. The discipline is yours.

- **Establish a baseline.** Know what correct looks like for this device, and keep an exported copy somewhere other than the device.
- **Snapshot before you change.** Named snapshot first, edits second. It costs fifteen seconds.
- **Preview every commit.** Especially the small ones. Nobody previews the change that breaks production.
- **Record intent.** Use the commit Description field — the log says what changed, only you can say why.
- **Export off-box.** A backup on the firewall is unavailable when the firewall is gone.
- **Verify after, not just before.** Check the System log for the commit result and the Configuration log for the change record.

Hands-On Lab 2: Managing Firewall Configurations

NETLAB+ · PAN v11 Firewall Essentials · load edu-210-11.0a-02.xml to start

- Load and commit a baseline configuration to the firewall.
- Save the configuration as a named snapshot, then export it to the client.
- Introduce a deliberate DNS error and undo it with Revert Changes.
- Modify SNMP settings and compare candidate to running with Preview Changes.
- Hide and reorder columns in the System log display.
- Create a severity filter by clicking, then combine two conditions.
- Use the Filter Builder to return System log entries from the last 60 minutes.



Credentials and addresses are in the Lab Settings table at the front of the lab guide.

Activity: Write the Change Record

Document the SNMP change from Lab 2 as if it were a production change request, then trade with a partner.



- State the change, the reason for it, and the expected result in one paragraph.
- Identify the rollback plan and the snapshot filename it depends on.
- Attach the Preview Changes output as evidence of what would be modified.
- Name the specific log entries that would confirm the change succeeded.
- Review a partner's record and list what is missing to reproduce or undo it.

The test: If your partner cannot undo your change from your record alone, the record is not finished.

Key Takeaways

-  Clicking OK edits the candidate configuration. Committing activates it. They are not the same act.
-  Save protects work from a reboot; export protects it from the firewall. Snapshot before you change.
-  Preview, Change Summary, and Validate each answer a different question. Use them before every commit.
-  Commit lock controls when a change goes live; config lock controls who can edit. Comment your locks.
-  The System log holds the result of your change; the Configuration log holds the record of it.

MID-STATE TECHNICAL COLLEGE · IT PROGRAMS

Questions?

Unit 3 · Managing Firewall Configurations

